

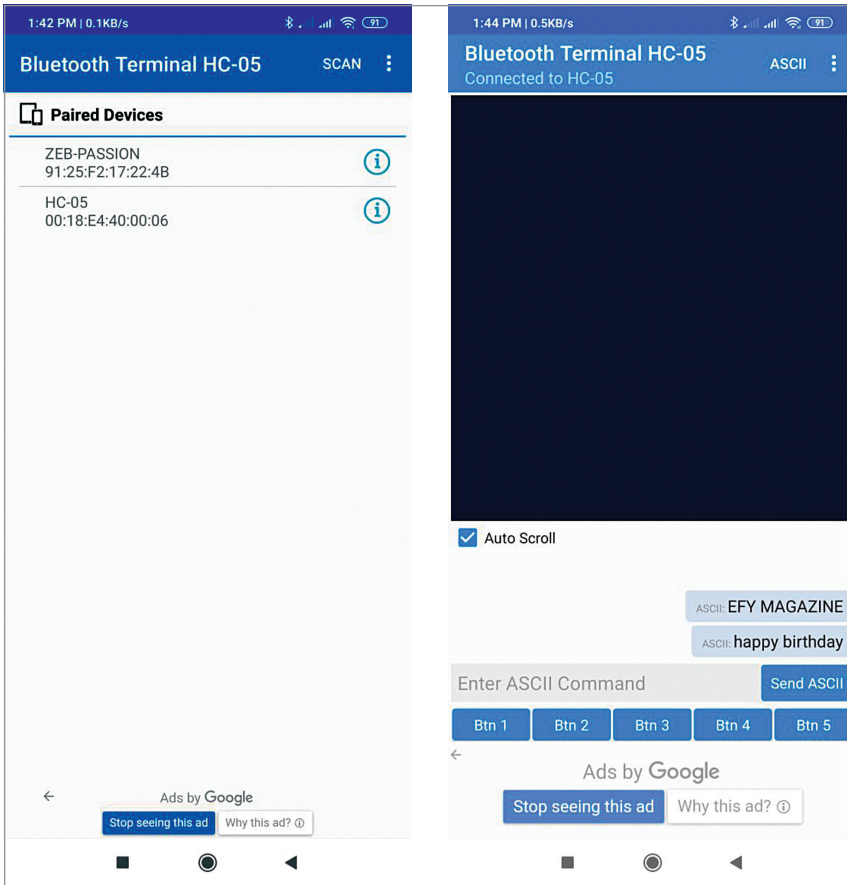


Fig. 8: Bluetooth terminal HC-05 app front view

Ride 7, PlatformIO + STM32, etc.

What has made this board popular is its ability to be programmed with the Arduino IDE. This way people can get started and build projects with STM32 in no time, since many are already familiar with Arduino IDE, which is an easy-to-use programming language and has readily available libraries. So, here we will be using the Arduino IDE to get started with STM32.

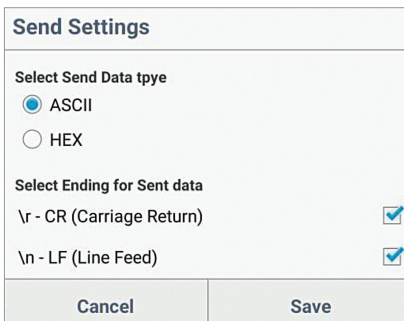


Fig. 9: \\r \\n Enable in ASCII mode

The STM32 Blue Pill development board does not come with a bootloader to make it Arduino IDE compatible. However, this bootloader can be flashed into the STM32 board and then the micro-USB

port can be directly used to upload the programs. Refer YouTube link below to learn how to load bootloader: <https://www.youtube.com/watch?v=Tm7IWQLrKYs&t=113s>

Based on this tutorial you should be able to easily upload bootloader software. After successful completion you can use the onboard micro USB port.

Follow the steps below to download and prepare the Arduino IDE for use with the STM32 board.

Step 1. If you have not yet installed the Arduino IDE, download and install it from the internet. Make sure you select your correct operating system.

Step 2. After Installing the Arduino IDE, open and download the required packages for the STM32 board. This can be done by selecting File→Preferences.

Step 3. Clicking on Preferences will open the dialogue box shown in Fig. 10. In the Additional Boards Manager URL text box, enter the link below and press OK:

http://dan.drown.org/stm32duino/package_STM32duino_index.json

Step 4. Go to Tool→Boards→Board Manager. This will open the Boards Manager dialogue box. Search for STM32F1 and install the package that appears.

Step 5. After the package instal-

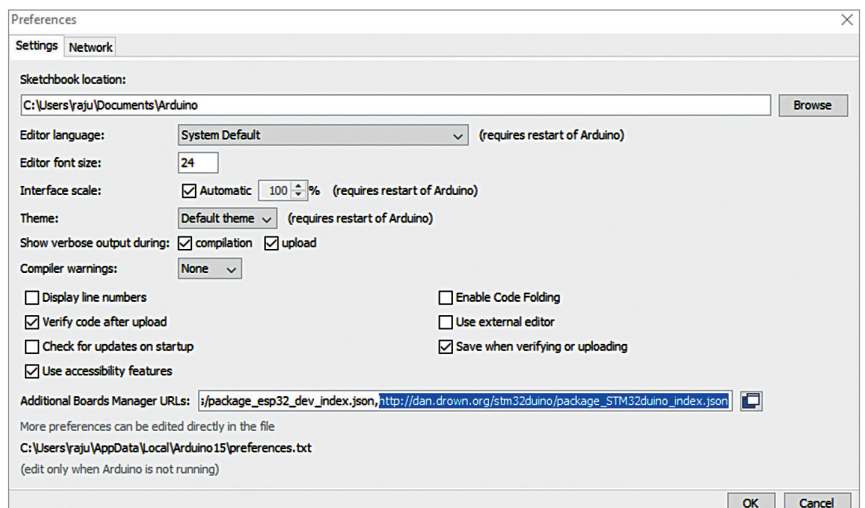


Fig. 10: Entering the link in Additional Boards Manager URLs